M13-C3 — Probe catalogue (H1–H7 + H4b) motivated by closed H0

H0 closed; H1–H5 run/analysed;
H6 run / analysed; H7 signposted only (not claims)

1 Status

Follow-up probes **motivated by closed baseline H0**, and **not** part of H0 itself. H0 is the completed six-cell AAMAS+ECAI investigation on fixture `m13_bucket3_binary_collider_and` / `n30_seed42`, documented in `learning_analysis.md` / `.tex` and summarised in `experiment.md`.

- H0: **closed** (see `learning_analysis.md`; not redefined here)
- H1: **run / analysed** — nested-prefix AAMAS ablation on `n25_seed42`. Record: `h1_support_ablation.md` / `.tex`. **Not a Bucket 3 claim.**
- H2: **run / analysed** — isolated-D AAMAS target c retains irrelevant d. Record: `h2_irrelevant_covariate.md` / `.tex`. **Not a Bucket 3 claim.**
- H3: **run / analysed** — BK-leading isolated a distracts ECAI target c from BD AND. Record: `h3_bk_leading_distractor.md` / `.tex`. **Not a Bucket 3 claim.**
- H4: **run / analysed** — repository `baseline_cautious` blocks brave root residual α-gadget on the H0 table. Record: `h4_cautious_vs_brave.md` / `.tex`. **Not a Bucket 3 claim.**
- H4b: **run / analysed** — on the H3 fixture, cautious changes only the α_3 close under shared leading-a gates (Layer B); Layer A shared with H3. Record: `h4b_cautious_split_under_a.md` / `.tex`. **Not H5. Not a Bucket 3 claim.**
- H5: **run / analysed** — experimental `greedy_cautious` matches AAMAS Greedy-brave on all 18 cells (Analysis A inert under Greedy). Record: `h5_greedy_cautious.md` / `.tex`. **Not a Bucket 3 claim. Not** an AAMAS-paper config.
- H6: **run / analysed** (`h6_folding_and_n_ablation.md`)
- H7: **signposted only** (prompted by H4 traces; not approved runs)
- **Not** approved Bucket 3 claims
- **Not** locked Bucket 1/2 claim content
- **Not** report-facing prose (`docs/report/findings/` is out of scope)
- Markdown twin: `future_probes.md`

Samuel decision (2026-08-01) — H1 / H2.

1. **H1 approved** — nested-prefix / missing-$(0, 0, 0)$ AAMAS ablation on this fixture (e.g. $n = 25$, seed 42, targets a and b) against the approved $n = 30$ baseline.
2. **H2 design approved** — separate schema-2 fixture with isolated $D \sim$ Bernoulli(1/2) (variables `a,b,c,d`).
3. **H2c in scope** for the first H2 write-up.

Samuel decision (2026-08-01) — H3.

1. Graph/naming sketch **approved**.
2. Root parameters **approved:** $B \sim$ Bernoulli(4/5) ($P(B{=}1) = 4/5$, $P(B{=}0) = 1/5$); $D \sim$ Bernoulli(7/10) ($P(D{=}1) = 7/10$, $P(D{=}0) = 3/10$); $A \sim$ Bernoulli(1/2) as previously sketched.
3. Fixture id **locked:** `m13_bucket3_binary_bd_and_lead_a` (AND of parents `b`,`d`; BK-leading isolated `a`). Distinct from `m13_bucket3_binary_collider_and` and from the completed H2 fixture id `m13_bucket3_binary_collider_and_iso_d`.
4. **AAMAS contrast in scope** for the H3 write-up (secondary to primary ECAI inspection of target c; valuable as a related but distinct check from H2).

Samuel decision (2026-08-03) — H4 / H5.

1. Add a repository-baseline cautious target-wise configuration, named `baseline_cautious`, which explicitly pins the existing engine defaults and keeps `check_ic` for output generation.
2. Permit explicit `brave` and `cautious` modes in the target-wise runner while leaving the published ECAI and AAMAS configurations unchanged.
3. Prepare H4 on the closed H0 fixture and sample, with all targets generated automatically; roots (a)/(b) are primary and child (c) is a control.
4. Do **not** run H4 as part of the infrastructure change.
5. Defer any cautious Greedy ABA Learning design to **H5**. No `aamas_cautious` configuration is approved.

H1–H5 have been run and analysed (`h1_support_ablation.md`, `h2_irrelevant_covariate.md`, `h3_bk_leading_distractor.md`, `h4_cautious_vs_brave.md`, `h4b_cautious_split_under_a.md`, `h5_greedy_cautious.md`). Approvals do not create a Bucket 3 claim or a run matrix.

2 Motivation (verified facts from the AAMAS arm of closed H0)

On the approved frozen sample `n30_seed42`, configuration `aamas2025` (greedy / `folding_selection(mgr)` / `folding_space(bk)` / brave / `check_ic`):

Target	Outcome	Delta (if any)
`c`	`solved`	`c(A) :- a_val_1(A), b_val_1(A).` (assumption-free)
`a`	`completed_no_solution`	empty
`b`	`completed_no_solution`	empty

Sources:

- collection summary: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/summary.md`
- target c delta: `.../cells/target-c/output/delta.aba`
- target a/b reports and `prolog.stdout` under `.../cells/target-{a,b}/`

For target c, the learned delta string coincides with the evaluator-only positive-state reference in `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/mechanism_reference.json`. Record that as **syntactic coincidence**, not as a causal-recovery verdict.

Procedural reading of the AAMAS root-target path (verified in `.../cells/target-a/output/prolog.stdout`; target b is symmetric):

1. Greedy folding builds **maximal co-occurrence bodies** from BK for each E^+ row (not a minimal-parent search).
2. The both-1 co-occurrence body `a(A) :- b_val_1(A), c_val_1(A)` is accepted under brave entailment.
3. The both-0 co-occurrence body `a(A) :- b_val_0(A), c_val_0(A)` over-generalises; assumption/contrary repair is attempted.
4. Contrary rote learning cites rows **26** and **30** — the only sample atoms $(0, 0, 0)$ in `samples/n30_seed42.csv` — and contrary folding then fails, yielding `completed_no_solution`.

These facts from closed H0 motivate probe families H1 and H2. They do not by themselves approve a Bucket 3 claim.

2.1 Motivation addendum (ECAI arm of closed H0 — for H3)

On the same frozen sample `n30_seed42`, configuration `ecai2024` (`folding_mode(nd)`, `folding_selection(any)`, brave / `check_ic`), target c **solved** with delta (verified):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

In `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout`, the first nd fold on an early E^+ row replaces the sample id by `a_val_1(A)` (the first BK predictor under declared order `a`, `b` for that cell), then assumption repair retains a in the ordinary target body while the second parent enters via a contrary. Collection summary: `.../ecai2024/n30_seed42/summary.md`.

This motivates **H3**: put an *independent* distractor first in BK by naming, under ECAI, on a new fixture. H3 is now **run / analysed** (`h3_bk_leading_distractor.md`); it is still **not** a Bucket 3 claim.

3 Ordering

1. **H1** — nested-prefix / missing-$(0, 0, 0)$ ablation on the existing fixture and AAMAS config (seed 42). **Run / analysed** (`h1_support_ablation.md`).
2. **H2** — separate schema-2 fixture with isolated covariate D on the $A \to C \leftarrow B$ AND collider. **Run / analysed** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains `d_val_*`.
3. **H3** — separate schema-2 fixture `m13_bucket3_binary_bd_and_lead_a` with BK-leading independent A and AND collider $B \to C \leftarrow D$. **Run / analysed** (`h3_bk_leading_distractor.md`). Lead: ECAI first-folds to a and retains a in the theory for c.
4. **H4** — same fixture/`n30_seed42`; locked ECAI brave vs repository `baseline_cautious`; roots a/b primary; c control. **Run / analysed** (`h4_cautious_vs_brave.md`). Lead: cautious KO on brave residual α-reuse; roots `completed_no_solution`; control c still solves.
5. **H5** — experimental `greedy_cautious` counterparts of all completed AAMAS collections. **Run / analysed** (`h5_greedy_cautious.md`). Lead: brave→cautious under Greedy is inert on these 18 cells.
6. **H6 (run / analysed); H7 (signposted only)** — H6 prompted by H4; H7 not approved runs.

Do not expand any family into a broader run matrix beyond these bounded probes without a further Samuel decision.

3.1 Relation among probe families (keep distinct)

Probe	Focus
H1	AAMAS roots; missing negative witness / both-0 over-generalisation on the **current** three-variable fixture
H2	AAMAS target c; irrelevant covariate enters **greedy maximal** co-occurrence bodies (minimality vs solvability); fixture id `m13_bucket3_binary_collider_and_iso_d`
H3	**ECAI** target c (primary); **BK order** puts independent distractor **first**; nd first-fold / repair path; **AAMAS contrast in scope**; id `m13_bucket3_binary_bd_and_lead_a`
H4	**Same fixture/sample**; locked ECAI brave vs repository `baseline_cautious`; roots a/b primary; c control; **run / analysed**
H5	Experimental `greedy_cautious` vs AAMAS brave on 18 cells; Analysis A inert; **run / analysed**; not AAMAS-paper config
H6	Run / analysed; length drivers under cautious+nd (H6a/H6b); brave `asm_intro(sechk)` vs `relto`

H3 is not a duplicate of H2: different DAG, different distractor placement/name, different primary config (ECAI nd vs AAMAS greedy), and distinct fixture ids. The in-scope AAMAS arm on H3 still differs from H2 because the irrelevant variable is BK-leading a and the mechanism parents are b, d.

H4 is not a duplicate of H1 or H3: it holds the table fixed and changes the learning mode while retaining the engine's baseline nd/any/all/relto options. H5 was kept separate because a cautious Greedy learner needed a paper- and algorithm-level specification before its experimental configuration could be interpreted.

4 Probe family H1 — support sensitivity on AAMAS root targets

Status: **run / analysed** / not a claim. Evidence record: `h1_support_ablation.md` / `h1_support_ablation.tex`.

Evaluative stance (locked for H1 reading). Roots have `no_observed_parent_deterministic_rule`. H0 AAMAS root `completed_no_solution` is the **desired / correct** outcome. H1 AAMAS root `solved` with co-occurrence Horn rules is an **incorrect / spurious** finite-sample solution.

4.1 H1a — spurious root rule under missing negative witness

On this fixture, AAMAS, target a (symmetrically b), seed 42: the nested prefix n=25 **omits all** $(0, 0, 0)$ **rows**. Learning **solved** with a delta that includes the both-0 co-occurrence rule

```
a(A) :- b_val_0(A), c_val_0(A).
```

(respectively `b(A) :- a_val_0(A), c_val_0(A).`), which is **false** as a population implication and is blocked on the closed H0 n=30 table by E^- rows 26 and 30.

Nested-prefix fact (seed 42): rows 1–25 contain no $(0, 0, 0)$ atom; rows 26 and 30 are the only $(0, 0, 0)$ observations in n=30. Verified: `n25_seed42.csv` is a row-for-row prefix of `n30_seed42.csv`.

4.2 H1b — negative witness in the unsafe cell

If the sample contains at least one opposite-label row in that both-0 predictor cell, the both-0 rule fails brave entailment under the observed AAMAS path. On closed H0 n=30 this manifested as `completed_no_solution`, not as emission of only the safe both-1 rule. That H0 outcome is the **correct** reading for roots under the H1 stance.

4.3 Explicit non-claims for H1

- Full support does **not** imply AAMAS will emit a "correct" deterministic root mechanism rule: roots have `no_observed_parent_deterministic_rule` in the evaluator reference.
- "One of each population atom" is a sufficient practical condition on this approved sample, but the operational point is a **label conflict inside the greedy candidate body cell**, not a generic demand for full joint support.
- H1 does not claim that omitting $(0, 0, 0)$ is the only way AAMAS can emit a spurious root rule, nor that every seed behaves identically.
- H1 does **not** claim that root `solved` on n=25 is progress or mechanism recovery.

4.4 Outcome (H1)

Run / analysed. Nested-prefix ablation executed on `m13_bucket3_binary_collider_and` / AAMAS / seed 42 / n=25. Roots a/b **incorrectly** solved with both-1 and both-0 co-occurrence rules. Child c remained AND-solved as a control. Full narrative: `h1_support_ablation.md`.

5 Probe family H2 — irrelevant isolated covariate on AAMAS target c

Status: **run / analysed** / not a claim. Evidence record: `h2_irrelevant_covariate.md` / `h2_irrelevant_covariate.tex`.

Lead finding. On fixture `m13_bucket3_binary_collider_and_iso_d` / `n30_seed42`, AAMAS solves target c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

so **irrelevant** D **is included**. The evaluator-only reference remains the D-free AND `c(A) :- a_val_1(A), b_val_1(A).`. Failure mode: distractor retention / minimality, not unlearnability.

5.1 H2a — irrelevant covariate enters greedy bodies

Observed as predicted: when both values of D appear among E^+ for c (9 each on n=30), greedy catalogue bodies **include** `d_val_*`.

5.2 H2b — learnability vs minimality

c **still solved** (two D-conditioned rules). Confirms that the interesting failure is retention of an irrelevant predictor, not `completed_no_solution`.

5.3 H2c — sample-edge (also run; not the centre)

Nested n=2 prefix where C=1 realises only d=1: single rule `c :- a_val_1, b_val_1, d_val_1`. Extreme edge. Documented briefly in `h2_irrelevant_covariate.md`. Not the scientific centre of H2.

5.4 Outcome (H2)

Run / analysed. Fixture authored and certified; AAMAS `n30_seed42` primary target c shows distractor inclusion; H2c n=2 also run. Full record: `h2_irrelevant_covariate.md`.

6 Probe family H3 — ECAI first-fold distraction by BK-leading independent variable

Status: **run / analysed** / not a claim. Evidence record: `h3_bk_leading_distractor.md` / `h3_bk_leading_distractor.tex`.

Lead finding. On fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, ECAI first-folds to `a_val_*` and retains a in both target rules for c. The emitted delta does **not** coincide with the evaluator-only BD AND `c(A) :- b_val_1(A), d_val_1(A).`. Failure mode: BK-leading distractor inclusion / first-fold distraction, not unlearnability.

6.1 Fixture (as run)

Field	Value
Fixture id	`m13_bucket3_binary_bd_and_lead_a`
Internal / display	`a`,`b`,`c`,`d` / A, B, C, D
DAG	$B \to C \leftarrow D$; **no edges** involving A
A	Bernoulli(1/2); independent of $\{B, C, D\}$
B	Bernoulli(4/5) — $P(B{=}0) = 1/5$, $P(B{=}1) = 4/5$
D	Bernoulli(7/10) — $P(D{=}0) = 3/10$, $P(D{=}1) = 7/10$
C	deterministic $C = B \wedge D$
Evaluator reference for c	`c(A) :- b_val_1(A), d_val_1(A).` (`a` must not appear)
BK order for target c	predictors `a`**, then** `b`**, then** `d` (irrelevant first)
Frozen sample	`n30_seed42`
Primary learning arm	**ECAI** target c
Secondary arm	**AAMAS** on the same frozen sample (**in scope**)

6.2 Scientific question (answered)

When learning target c under **ECAI** on this frozen sample, does BK-leading independent a **distract** the first fold and/or the final ABA shape away from the mechanism-aligned B, D conjunction?

Answer (bounded): yes — first fold selects `a_val_1`; both target rules retain a; metrics body vars $\{a\}$. Secondary: AAMAS also retains `a_val_*` in both Horn bodies (vs H2's trailing-d inclusion).

6.3 Secondary finding (in scope; not the lead)

Brave residual / underdetermining delta on the nested a=1 / α_1–α_3 side. Cross-link H0 ECAI target-a rows 9 vs 26. Reinforces: `solved` + **audit SAT** $\neq$ **mechanism-aligned**.

6.4 Explicit non-claims for H3

- Inclusion of a is **not** causal recovery of an $A \to C$ edge.
- Runner `solved` is **not** automatically correct / mechanism-aligned.
- H3 does not reuse H2's graph; distractor here is A, parents B, D.
- Completing H3 does **not** approve a Bucket 3 claim.

6.5 Outcome (H3)

Run / analysed. Fixture authored and certified; ECAI `n30_seed42` primary target c shows BK-leading a distraction; AAMAS contrast also retains a. Full record: `h3_bk_leading_distractor.md`.

7 Probe family H4 — repository-baseline cautious vs published ECAI brave

Status: **run** / **analysed** / not a claim. Evidence record: `h4_cautious_vs_brave.md` / `h4_cautious_vs_brave.tex`.

Lead finding. On the closed H0 table `m13_bucket3_binary_collider_and` / `n30_seed42`, switching from locked **ECAI brave** to repository `baseline_cautious` **blocks the brave residual-assumption gadget** that had allowed roots a/b to `solved`. Paths coincide through early α_1–α_3 setup; at the closing step `c_alpha_3 :- alpha_2, ...` brave says **OK** and finishes, cautious says **KO**, then exhausts the folding budget and returns `completed_no_solution` (no delta). Control c still solves with a byte-identical delta to locked ECAI c.

7.1 Fixture / arms (as run)

Field	Value
Fixture / sample	`m13_bucket3_binary_collider_and` / `n30_seed42`
Sample hash	`sha256:1edae465...a8ca`
Brave arm	locked `ecai2024`
Cautious arm	`baseline_cautious` (`configs/baseline_cautious_config.pl`)
Primary	roots a, b
Control	c

7.2 Scientific question (answered)

Does repository-baseline cautious learning change the ECAI-brave root outcomes that relied on a residual per-row assumption gadget (rows 9 vs 26), and what happens to control c?

Answer (bounded): yes for roots — cautious KO at the brave closing α-reuse; `completed_no_solution`. Control c still `solved` with identical delta.

7.3 Explicit non-claims for H4

- Not claimed that cautious mode is "better" for causal recovery.
- Not claimed that cautious root no-solution recovers a correct root mechanism (none exists).
- Not a ranking of brave vs cautious as general ABALearn policy.
- Control c string match / `solved` is not causal recovery.
- Under cautious learning, SAT of `bk.sol_chk.asp` is an existential final-artefact integrity witness, not a cautious-consequence proof.

7.4 Outcome (H4)

Run / analysed. Collection at `.../baseline_cautious/n30_seed42/`: $a/b =$ `completed_no_solution`; $c =$ `solved` (identical delta to ECAI). Full record: `h4_cautious_vs_brave.md`.

7.5 Prompted follow-ups (signpost only; not H4 evidence)

H5 is now **run / analysed** separately (`h5_greedy_cautious.md`). H6 is now **run / analysed** (`h6_folding_and_n_ablation.md`). H7 (brave `asm_intro(sechk)` vs `relto` on the 9/26 conflict) remains a future probe prompted by H4 traces — not started.

8 Probe family H4b — cautious α_3 close under H3 leading-a gates

Status: **run / analysed** / not a claim. **Not H5.** Evidence record: `h4b_cautious_split_under_a.md` / `h4b_cautious_split_under_a.tex`.

Lead finding (Layer B). On H3 fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42` target c, repository `baseline_cautious` keeps H3's leading-a gates and the persistent α_1-nest vs α_2-flat contrary split; it **only** changes the α_3 close — rejecting brave residual `c_alpha_3 :- alpha_1, a_val_1` and accepting `c_alpha_3 :- d_val_1`. Both arms `solved`.

Layer A (shared with H3; not a cautious effect): a-gating of target rules; nested `c_alpha_1` vs flat `c_alpha_2`; procedural reason is search order / `relto` / cover-and-repair (see the H4b record).

8.1 Outcome (H4b)

Run / analysed. Collection: `.../m13_bucket3_binary_bd_and_lead_a/baseline_cautious/n30_seed42/`; target c `solved` ($\sim$2.9s; audit SAT). Incidental: $a/b/d$ timed out (outside focus). Full record: `h4b_cautious_split_under_a.md`.

9 Probe family H5 — experimental Greedy + cautious (`greedy_cautious`)

Status: **run / analysed** / not a claim. Evidence record: `h5_greedy_cautious.md` / `h5_greedy_cautious.tex`.

Lead finding (Analysis A). On all **18** paired cells, experimental `greedy_cautious` matches locked `aamas2025` on every outcome and every solved delta (byte-identical). Traces are isomorphic aside from entailment-logging surface differences. Flipping brave→cautious inside the Greedy bundle does not change accepted or rejected theories on these tables. Mean runtime $\approx 1.84\times$ (still sub-second).

Secondary (Analysis B). Large contrasts appear only vs repository `baseline_cautious` (nd bundle) on H0/H3 n=30 — **search-strategy**, not the brave→cautious flip. Not a single-knob `folding_mode` ablation.

Config: `configs/greedy_cautious_config.pl` (experimental; not published AAMAS; do not call it `aamas_cautious`).

9.1 Outcome (H5)

Run / **analysed.** Full narrative: `h5_greedy_cautious.md`.

10 Probe family H6 — folding-steps and nested-n search-cost ablation

Status: **run** / **analysed** / not a claim. Evidence: `h6_folding_and_n_ablation.md` / `.tex`.

H6a: fixed `n30_seed42`; `folding_steps`$(M) \in \{1, 2, 5, 10\}$. Root trace length $\approx$ linear in M via nd failed-band replays (`NEW` $= 9 \times M$); control c solves inside `tokens`(1).

H6b: fixed M=2; nested $n \in \{30, 60, 90\}$. Topology fixed (NEW=18, KO=32); length $\approx$ linear in n via per-band bookkeeping.

Cross: H4's ∼5k-line root no-sols are primarily cumulative token-budget replays (H6a) with secondary n-scaling at fixed M=2 (H6b). Not a claim; not an $n\times$steps interaction result.

10.1 Outcome (H6)

Run / **analysed.** Full narrative: `h6_folding_and_n_ablation.md`.

11 Boundary distinctions to preserve

Keep separate throughout any future probe write-up:

Object	Role
AAMAS strategy / greedy maximal co-occurrence search	procedural learner behaviour (H1/H2)
ECAI nd first-fold / BK serialisation order	procedural learner behaviour (H3)
Brave vs cautious entailment / learning mode	procedural / semantic learner behaviour (H4)
Cautious Greedy specification / experimental `greedy_cautious`	procedural / semantic probe (H5); not an AAMAS-paper config
Finite-sample support and label conflicts in candidate body cells	sample information
Mechanism correspondence for c (AND)	evaluator-only interpretation of a non-root
Absence of deterministic root mechanisms	evaluator reference status
Syntactic delta = reference string	coincidence, not automatic causal verdict
Runner `solved` under brave	brave coverage of $E^{\pm}$, not mechanism recovery

Do not treat predictive rules for roots as mechanism recovery.

12 Cross-links

- Investigation record: `experiment.md`
- Mathematical fixture dossier: `fixture_dossier.tex`
- Bucket 3 planning hub: `../M1.3-bucket3-claims.md` (still **no claim**)
- Research decision log: `docs/research/decisions.md` (2026-08-01 H1–H3 entries; 2026-08-03 H4/H5 entry)
- Fixture pre-run bundle: `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/`
- AAMAS baseline: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/`
- ECAI baseline (H3/H4 motivation): `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/ecai2024/n30_seed42/`
- Repository cautious baseline: `configs/baseline_cautious_config.pl`
- H4 config: `causal/configs/targetwise/m13_bucket3_binary_collider_and/ baseline_cautious/n30_seed42.yaml`
- H4 collection: `causal/outputs/aba_learning/targetwise/ m13_bucket3_binary_collider_and/baseline_cautious/n30_seed42/`
- Target-wise explicit brave/cautious validator: `causal/targetwise/config.py`
- Entailment implementation: `asp_engine.pl` (`entails/5`)

13 Decisions recorded

Item	Decision	Date
H1 nested-prefix / missing-$(0, 0, 0)$ AAMAS ablation	**Approved** then **run / analysed**	2026-08-01 / 2026-08-02
H2 four-variable isolated-D fixture + AAMAS c	**Approved** then **run / analysed**	2026-08-01 / 2026-08-02
H2c in first H2 write-up	**In scope** (run; brief note only)	2026-08-01 / 2026-08-02
H3 graph/naming; B 80/20, D 70/30; id `m13_bucket3_binary_bd_and_lead_a`; AAMAS contrast	**Design approved** then **run / analysed**	2026-08-01 / 2026-08-02
H4 repository-baseline cautious vs published ECAI brave	**Infrastructure ready** then **run / analysed**	2026-08-03
H4b cautious α_3 close on H3 leading-a target c	**Run / analysed** (not H5)	2026-08-03
H5 experimental `greedy_cautious` (18 cells)	**Run / analysed**	2026-08-03
H6 folding/n ablation (H4-prompted)	**Approved** then **run / analysed**	2026-08-03
H7 (H4-prompted)	**Signposted only; not started**	2026-08-03

14 Current status and remaining work

- **H1: complete** (`h1_support_ablation.md`). Not a claim.
- **H2: complete** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains irrelevant d. Not a claim.
- **H3: complete** (`h3_bk_leading_distractor.md`). Lead: ECAI c first-folds to / retains BK-leading a (not BD AND). Not a claim.
- **H4: complete** (`h4_cautious_vs_brave.md`). Lead: cautious blocks brave root residual α-gadget; roots `completed_no_solution`; control c still solves. Not a claim.
- **H4b: complete** (`h4b_cautious_split_under_a.md`). Lead (Layer B): only `c_alpha_3` close changes under shared Layer A. Not a claim; not H5.
- **H5: complete** (`h5_greedy_cautious.md`). Lead: Greedy brave→cautious inert on 18 cells; Analysis B contrasts are search-bundle. Not a claim; not AAMAS-paper config.
- **H6: complete** (`h6_folding_and_n_ablation.md`). Lead: root cost $\approx$ linear in `folding_steps`(M) and nested n at $M{=}2$. Not a claim.
- **H7:** signposted only; not started.
- Still **no** Bucket 3 claim and **no** expanded run matrix without a further Samuel decision.